

PSP data to downstream

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Context

Multiple downstream get data from PSP database and store in data lake and warehouse. From lake, data gets consumed by analytical team to generate reports/do analytics on PSP data.

The aim of this document is to understand this process and optimize to serve business use cases from analytical/internal teams.

Business Use-Case

- Analytical team need psp data to run analytics
- Internal team need data to create product specific/feature specific dashboards

Current state of data flow

UIP

Data ingestion to lake via data pull method by connecting to PSP read replica database.

Job frequency

daily

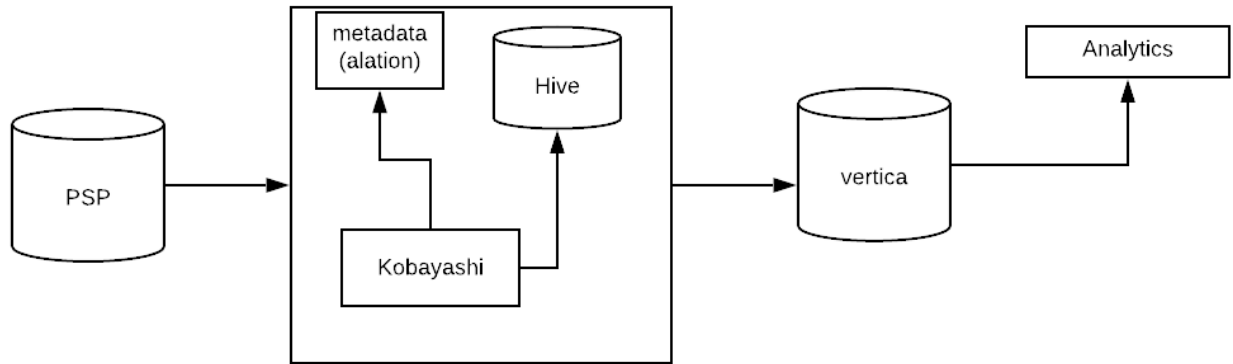
Job duration

~220 mins

Handle DDL changes at source

DDL changes at source will get reflected and data ingestion starts accordingly.

how it works



Limitations

- Currently not all PSP tables are getting ingested into lake.
- Incremental load type is based on CDC column (modified_date) which is not true for all source tables.
- Source/analytical team need to raise requests for any new table ingestion.

PSP Tables

Catalog has all tables but currently only below tables are getting ingested into lake

S.No.	Table Name	Refresh type	Data Lake Schema Name	Remarks
1	PSP_ADDRESS	INCREMENTAL	UED_PSP_DWH	Table names and column names are same in data lake schema.
2	PSP_AUTH_USER	INCREMENTAL	UED_PSP_DWH	
3	PSP_BANK_ACCOUNT	INCREMENTAL	UED_PSP_DWH	
4	PSP_BILL	INCREMENTAL	UED_PSP_DWH	
5	PSP_BILL_PAYMENT	INCREMENTAL	UED_PSP_DWH	
6	PSP_BILL_PAYMENT_SPLIT	INCREMENTAL	UED_PSP_DWH	
7	PSP_BILLING_DETAIL	INCREMENTAL	UED_PSP_DWH	
8	PSP_COMPANY	INCREMENTAL	UED_PSP_DWH	
9	PSP_COMPANY_AGENCY	INCREMENTAL	UED_PSP_DWH	
10	PSP_COMPANY_BANK_ACCOUNT	INCREMENTAL	UED_PSP_DWH	
11	PSP_COMPANY_EVENT	INCREMENTAL	UED_PSP_DWH	
12	PSP_COMPANY_EVENT_DETAIL	INCREMENTAL	UED_PSP_DWH	
13	PSP_COMPANY_LAW	INCREMENTAL	UED_PSP_DWH	
14	PSP_COMPANY_OFFER	INCREMENTAL	UED_PSP_DWH	
15	PSP_COMPANY_OFFERING	INCREMENTAL	UED_PSP_DWH	
16	PSP_COMPANY_PAYROLL_ITEM	INCREMENTAL	UED_PSP_DWH	
17	PSP_COMPANY_SERVICE	INCREMENTAL	UED_PSP_DWH	
18	PSP_COMPANY_USAGE	INCREMENTAL	UED_PSP_DWH	
19	PSP_COMPENSATION	INCREMENTAL	UED_PSP_DWH	
20	PSP_CONTACT	FULL_REFRESH	UED_PSP_DWH	
21	PSP_DDCOMPANY_SERVICE_INFO	FULL_REFRESH	UED_PSP_DWH	
22	PSP_DEDUCTION	INCREMENTAL	UED_PSP_DWH	

23	PSP_EMPLOYEE	INCREMENTAL	UED_PSP_DWH	
24	PSP_EMPLOYEE_BANK_ACCOUNT	INCREMENTAL	UED_PSP_DWH	
25	PSP_ENTITLEMENT	INCREMENTAL	UED_PSP_DWH	
26	PSP_ENTITLEMENT_CODE	INCREMENTAL	UED_PSP_DWH	
27	PSP_ENTITLEMENT_UNIT	INCREMENTAL	UED_PSP_DWH	
28	PSP_FINANCIAL_TRANS_STATE	INCREMENTAL	UED_PSP_DWH	
29	PSP_FINANCIAL_TRANSACTION	INCREMENTAL	UED_PSP_DWH	
30	PSP_FRAUD_RULE	INCREMENTAL	UED_PSP_DWH	
31	PSP_GEMS_LEDGER_POSTING_RULE	INCREMENTAL	UED_PSP_DWH	
32	PSP_GEMS_MONTHLY_BALANCE	INCREMENTAL	UED_PSP_DWH	
33	PSP_GEMS_UPLOAD_BATCH	INCREMENTAL	UED_PSP_DWH	
34	PSP_INDIVIDUAL	INCREMENTAL	UED_PSP_DWH	
35	PSP_LEDGER_BALANCE	INCREMENTAL	UED_PSP_DWH	
36	PSP_LIMIT_RULE	INCREMENTAL	UED_PSP_DWH	
37	PSP_MONEY_MOVEMENT_TRANSACTION	INCREMENTAL	UED_PSP_DWH	
38	PSP_OFFER	INCREMENTAL	UED_PSP_DWH	
39	PSP_OFFERING	INCREMENTAL	UED_PSP_DWH	
40	PSP_OFFERING_SVCCHG	INCREMENTAL	UED_PSP_DWH	
41	PSP_OFFERING_SVCCHG_GRP	INCREMENTAL	UED_PSP_DWH	
42	PSP_ON_HOLD_REASON	INCREMENTAL	UED_PSP_DWH	
43	PSP_PAYCHECK	INCREMENTAL	UED_PSP_DWH	
44	PSP_PAYCHECK_SPLIT	INCREMENTAL	UED_PSP_DWH	
45	PSP_PAYCHECK_USAGE	INCREMENTAL	UED_PSP_DWH	
46	PSP_PAYEE	INCREMENTAL	UED_PSP_DWH	
47	PSP_PAYROLL_FREQUENCY	FULL_REFRESH	UED_PSP_DWH	
48	PSP_PAYROLL_RUN	INCREMENTAL	UED_PSP_DWH	
49	PSP_POSTING_RULE	FULL_REFRESH	UED_PSP_DWH	
50	PSP_QUICKBOOKS_INFO	INCREMENTAL	UED_PSP_DWH	
51	PSP_SERVICE_STATUS	FULL_REFRESH	UED_PSP_DWH	
52	PSP_SERVICE_SUB_STATUS	FULL_REFRESH	UED_PSP_DWH	
53	PSP_TRANSACTION_RESPONSE	INCREMENTAL	UED_PSP_DWH	
54	PSP_TRANSACTION_RETURN	INCREMENTAL	UED_PSP_DWH	
55	PSP_TRANSACTION_STATE	FULL_REFRESH	UED_PSP_DWH	
56	PSP_TRANSACTION_TYPE	FULL_REFRESH	UED_PSP_DWH	
57	PSP_USAGE_PERIOD	INCREMENTAL	UED_PSP_DWH	
58	PSP_VMP_EMPLOYEE_INFO	FULL_REFRESH	UED_PSP_DWH	

EDP

data ingestion to warehouse via data pull method by connecting to PSP read replica database

Job frequency

daily frequency

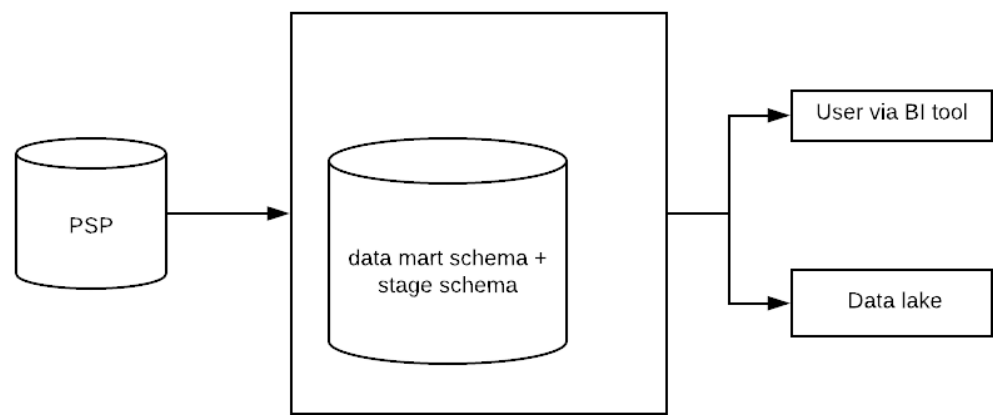
Job duration

~90 mins

Handle DDL changes at source

Based on request from further downstream or source.

how it works



Limitations

- Data from PSP tables are getting pulled for certain use-cases. Not generic.
- Incremental load type is based on CDC column (modified_date) which is not true for all source tables.

PSP tables

S.No.	Table Name	Refresh type
1	PSP_ADDRESS	INCREMENTAL
2	PSP_AUTH_USER	INCREMENTAL
3	PSP_BANK_ACCOUNT	INCREMENTAL
4	PSP_BATCH_JOB_AUDIT_LOG	INCREMENTAL
5	PSP_BILL	INCREMENTAL
6	PSP_BILL_PAYMENT	INCREMENTAL
7	PSP_BILL_PAYMENT_SPLIT	INCREMENTAL
8	PSP_BILLING_DETAIL	INCREMENTAL
9	PSP_BILLING_DETAIL_ARCHIVE	INCREMENTAL
10	PSP_COMP_ADJUST_SUBMISSION	INCREMENTAL
11	PSP_COMP_PMT_TEMPLATE_AGENCYID	INCREMENTAL
12	PSP_COMP_PMTTEMPLATE_PMTMETHOD	INCREMENTAL
13	PSP_COMPANY	INCREMENTAL
14	PSP_COMPANY_AGENCY	INCREMENTAL
15	PSP_COMPANY_BANK_ACCOUNT	INCREMENTAL
16	PSP_COMPANY_EVENT	INCREMENTAL
17	PSP_COMPANY_EVENT_ARC	INCREMENTAL
18	PSP_COMPANY_EVENT_DETAIL	INCREMENTAL
19	PSP_COMPANY_LAW	INCREMENTAL
20	PSP_COMPANY_LAW_RATE	INCREMENTAL
21	PSP_COMPANY_OFFER	INCREMENTAL
22	PSP_COMPANY_OFFERING	INCREMENTAL

23	PSP_COMPANY_SERVICE	INCREMENTAL
24	PSP_COMPANY_USAGE	INCREMENTAL
25	PSP_COMPANYAGENCY_PMTTEMPLATE	INCREMENTAL
26	PSP_CONTACT	INCREMENTAL
27	PSP_DDCOMPANY_SERVICE_INFO	INCREMENTAL
28	PSP_DELETED_RECORD	INCREMENTAL
29	PSP_EFFECTIVE_DEPOSIT_FREQ	INCREMENTAL
30	PSP_EFTPS_PAYMENT_DETAIL	INCREMENTAL
31	PSP_EMPLOYEE	INCREMENTAL
32	PSP_EMPLOYEE_ACCRUAL	INCREMENTAL
33	PSP_EMPLOYEE_BANK_ACCOUNT	INCREMENTAL
34	PSP_EMPLOYEE_USAGE	INCREMENTAL
35	PSP_ENTITLEMENT	INCREMENTAL
36	PSP_ENTITLEMENT_CODE	INCREMENTAL
37	PSP_ENTITLEMENT_UNIT	INCREMENTAL
38	PSP_ENTITLEMENT_UPD	INCREMENTAL
39	PSP_FINANCIAL_TRANS_STATE	INCREMENTAL
40	PSP_FINANCIAL_TRANSACTION	INCREMENTAL
41	PSP_INDIVIDUAL	INCREMENTAL
42	PSP_LAW	INCREMENTAL
43	PSP_LEDGER_ACCOUNT	INCREMENTAL
44	PSP_LEDGER_BALANCE	INCREMENTAL
45	PSP_LIABILITY_ADJUSTMENT	INCREMENTAL
46	PSP_MONEY_MOVEMENT_TRANSACTION	INCREMENTAL
47	PSP_OFFER	INCREMENTAL
48	PSP_OFFERING	INCREMENTAL
49	PSP_ON_HOLD_REASON	INCREMENTAL
50	PSP_PAYCHECK	INCREMENTAL
51	PSP_PAYCHECK_ACCRUAL	INCREMENTAL
52	PSP_PAYCHECK_ARCHIVE	INCREMENTAL
53	PSP_PAYCHECK_SPLIT	INCREMENTAL
54	PSP_PAYCHECK_USAGE	INCREMENTAL
55	PSP_PAYEE	INCREMENTAL
56	PSP_PAYROLL_FREQUENCY	INCREMENTAL
57	PSP_PAYROLL_RUN	INCREMENTAL
58	PSP_PAYROLL_SUBTYPE	INCREMENTAL
59	PSP_PMT_TEMPLATE_FREQUENCY	INCREMENTAL
60	PSP_PMT_TEMPLATE_PAYMENTMETHOD	INCREMENTAL
61	PSP_PMTTEMPLATE_PRINTEDCHKINFO	INCREMENTAL
62	PSP_QBDT_PAYCHECK_INFO	INCREMENTAL
63	PSP_QUICKBOOKS_INFO	INCREMENTAL

64	PSP_SERVICE	INCREMENTAL
65	PSP_SERVICE_STATUS	INCREMENTAL
66	PSP_SERVICE_SUB_STATUS	INCREMENTAL
67	PSP_SVCSTAT_SVC_ASSOC	INCREMENTAL
68	PSP_TAX	INCREMENTAL
69	PSP_TRANSACTION_RETURN	INCREMENTAL
70	PSP_USAGE_PERIOD	INCREMENTAL
71	PSP_VOIDED_CHECK	INCREMENTAL

Business Objects

Pulls data by connecting to PSP read replica timely and generate reports.

Job frequency

TBD

Job duration

TBD

Handle DDL changes at source

TBD

how it works

TBD

Limitations

TBD.

PSP tables

TBD

Target state

UIP

Data will be pushed to IAC data lake via goldengate trail files. Hence lag to data lake would be reduced and no need to connect to read replica database.

PSP tables

All tables under application schema from **monolith** and **IBOB** database

Solution to multiple use cases

This could be solution to lot of problems which needs PSP data without connecting to databases and almost real time. For this, an individual might need to understand process, connect to data lake and consume accordingly.

- Analytical team to run analytics and derive data insights for product
- Internal or PD team to create product specific/feature dashboards

UED_PSP_DWHUED_PSP_DWH